

BEAUSOLEIL, ON, Canada

WMO: 714325

Lat: 44.850N

Lon: 79.870W

Elev: 600

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-5.7	-18.5	2.0	-10.6	-12.2	2.9	-5.0	16.4	20.9	15.0	20.5	3.5	340	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	85.9	74.0	82.6	71.7	79.7	70.3	75.9	82.6	73.9	79.9	72.0	77.2	5.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	129.2	79.5	71.9	120.8	77.2	70.2	113.8	75.2	39.8	82.8	37.9	80.6	36.2	77.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	12.1	10.7	-18.5	91.0	8.5	3.2	-24.6	93.3	-29.6	95.2	-34.4	97.0	-40.5	99.3
			-20.0	79.5	8.3	2.2	-25.9	81.1	-30.8	82.4	-35.5	83.7	-41.5	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	17.7	18.9	27.8	41.1	53.2	63.9	68.8	67.7	61.0	48.3	37.7	26.7
	DBStd	20.06	13.49	10.47	10.22	8.56	8.10	7.33	5.62	5.49	7.16	8.34	8.69	10.22
	HDD50	4151	1002	871	691	288	59	2	0	0	6	134	374	724
	HDD65	7855	1467	1291	1154	717	375	107	27	34	157	520	818	1188
	CDD50	2155	0	0	2	21	159	418	582	548	336	82	6	1
	CDD65	387	0	0	0	1	10	74	145	117	37	3	0	0
	CDH74	2459	0	0	0	16	95	548	899	661	220	20	0	0
	CDH80	595	0	0	0	2	20	150	208	174	38	3	0	0
Wind	WSAvg	4.8	5.4	5.1	5.0	5.3	4.7	4.0	3.8	3.8	4.2	4.8	5.6	5.6
Precipitation	PrecAvg	36.3	3.7	2.3	2.2	2.6	3.2	3.1	3.3	2.4	3.3	3.3	3.5	3.7
	PrecMax	40.7	5.9	4.4	3.8	5.3	6.5	6.3	6.1	4.4	5.0	5.9	6.5	5.6
	PrecMin	29.2	2.0	0.8	0.6	0.8	1.3	0.9	1.1	0.7	1.0	0.8	1.3	1.5
	PrecStd	3.2	1.1	0.9	0.9	1.2	1.1	1.2	1.2	0.9	1.1	1.5	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	47.0	61.4	75.6	82.3	88.6	89.0	90.6	85.2	76.5	61.4	52.8
		MCWB	44.6	42.3	55.8	61.5	68.4	75.5	77.3	74.7	71.5	65.3	56.4	49.9
	2%	DB	41.7	40.9	51.2	65.7	75.8	84.4	85.0	84.5	79.9	69.5	56.7	47.0
		MCWB	39.9	37.7	47.6	54.1	65.2	73.0	74.3	73.2	69.7	63.4	53.0	44.7
	5%	DB	38.2	37.0	45.2	59.6	70.8	80.5	82.0	80.6	76.2	64.5	52.6	42.7
		MCWB	36.8	35.0	40.8	50.7	61.6	71.4	71.8	71.1	67.2	59.7	49.5	40.5
	10%	DB	34.7	34.1	40.8	53.9	66.3	76.2	79.2	77.1	72.5	60.7	49.4	38.9
		MCWB	33.6	32.6	37.9	46.5	59.1	68.6	70.4	68.9	65.8	56.6	47.0	37.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	42.8	57.0	62.9	71.2	77.4	79.3	78.2	73.7	68.4	58.2	51.2
		MCDB	46.1	44.6	60.0	70.6	77.4	84.5	86.0	83.7	81.5	73.4	60.6	53.2
	2%	WB	40.2	38.5	47.8	56.8	67.2	74.7	76.3	74.7	71.1	63.8	53.9	45.0
		MCDB	41.1	39.9	51.6	64.7	74.0	82.1	82.3	81.5	76.7	67.6	56.1	46.8
	5%	WB	36.8	35.2	41.6	51.5	63.6	72.5	74.1	72.7	69.0	60.3	50.0	40.7
		MCDB	38.1	36.8	44.1	57.9	68.1	78.8	79.3	78.0	74.5	64.2	52.0	41.9
	10%	WB	33.8	32.7	38.1	47.4	60.2	69.9	72.1	70.9	66.5	57.0	47.0	37.4
		MCDB	34.7	34.1	40.4	52.7	65.4	74.7	76.8	75.2	71.2	59.9	49.0	38.8
Mean Daily Temperature Range	5% DB	MDBR	15.9	18.5	17.2	17.8	18.4	18.2	17.3	16.7	16.7	14.4	12.3	12.7
		MCDBR	14.1	18.6	20.4	27.4	25.1	23.2	20.3	20.9	21.5	20.8	17.4	15.0
		MCWBR	12.9	16.0	15.5	17.4	14.7	12.9	10.8	10.1	11.6	13.7	14.6	12.5
	5% WB	MCDBR	13.9	16.7	18.9	25.2	22.2	21.3	18.0	17.3	19.0	19.6	16.3	13.7
		MCWBR	12.9	14.7	15.6	17.7	14.6	13.1	11.0	9.9	11.8	13.9	14.7	12.3
Clear-Sky Solar Irradiance	taub	0.287	0.296	0.317	0.352	0.384	0.405	0.408	0.401	0.365	0.345	0.313	0.286	
	taud	2.396	2.338	2.387	2.369	2.350	2.311	2.329	2.340	2.431	2.464	2.499	2.458	
	Ebn at Noon	264	283	292	288	281	274	272	270	272	262	253	253	
	Edh at Noon	23	30	33	36	38	40	39	37	31	26	22	20	
All-Sky Solar Radiation	RadAvg	412	698	1128	1433	1705	1862	1884	1614	1261	736	425	297	
	RadStd	49	87	118	133	167	140	118	88	121	81	60	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page